Predicting Lap Times from Telemetry Data

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CSE 4224 – Introduction to Machine Learning
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Project Goals

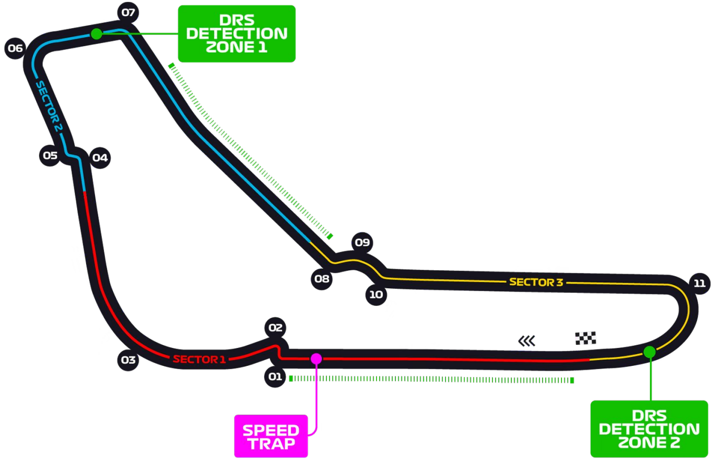
Goals

- ➢ Predict Lap Times for a Specific Driver (me)
- ➢ Understand which driving inputs/features impact lap times the most
- ➢ Compare Model Effectiveness
    - ○ Linear Regression vs. Decision Trees
- ➢ Learn how to improve my models

Project Data

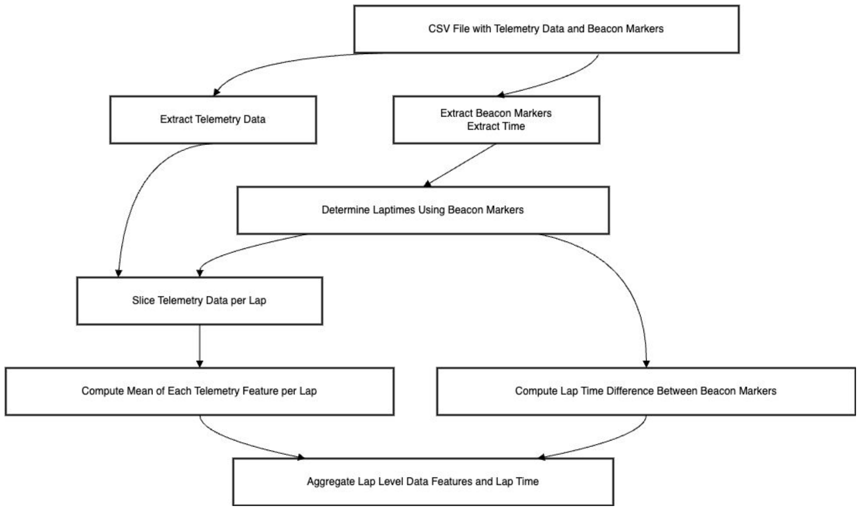
Data Collection

- Game: Assetto Corsa Competizione
- Track: Monza
- Car: Porsche 991ii GT3 R EVO
- Logged telemetry data using MoTeC I2
- Features consist of speed, throttle, brakes, suspension, etc



Data Preparation

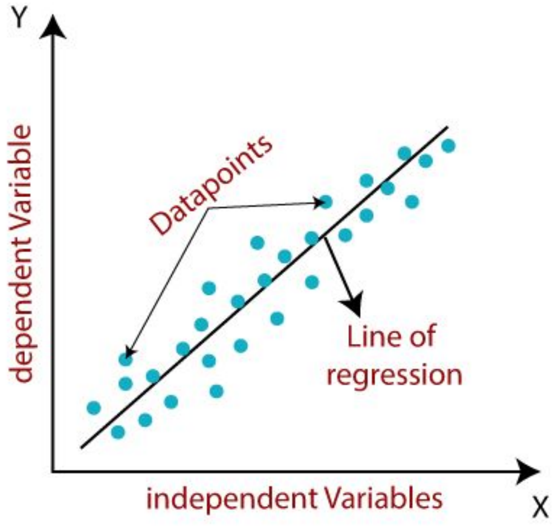
- ➢ List of “Beacon Markers” which were used to get the lap times
  - 201.989 316.961 430.743 544.987 …
- ➢ Aggregated laps and telemetry data using the beacon markers
- ➢ Converted to numerical data
- ➢ Removed major outliers and missing data
  - I kept one crash in because I was curious to see how the models would react



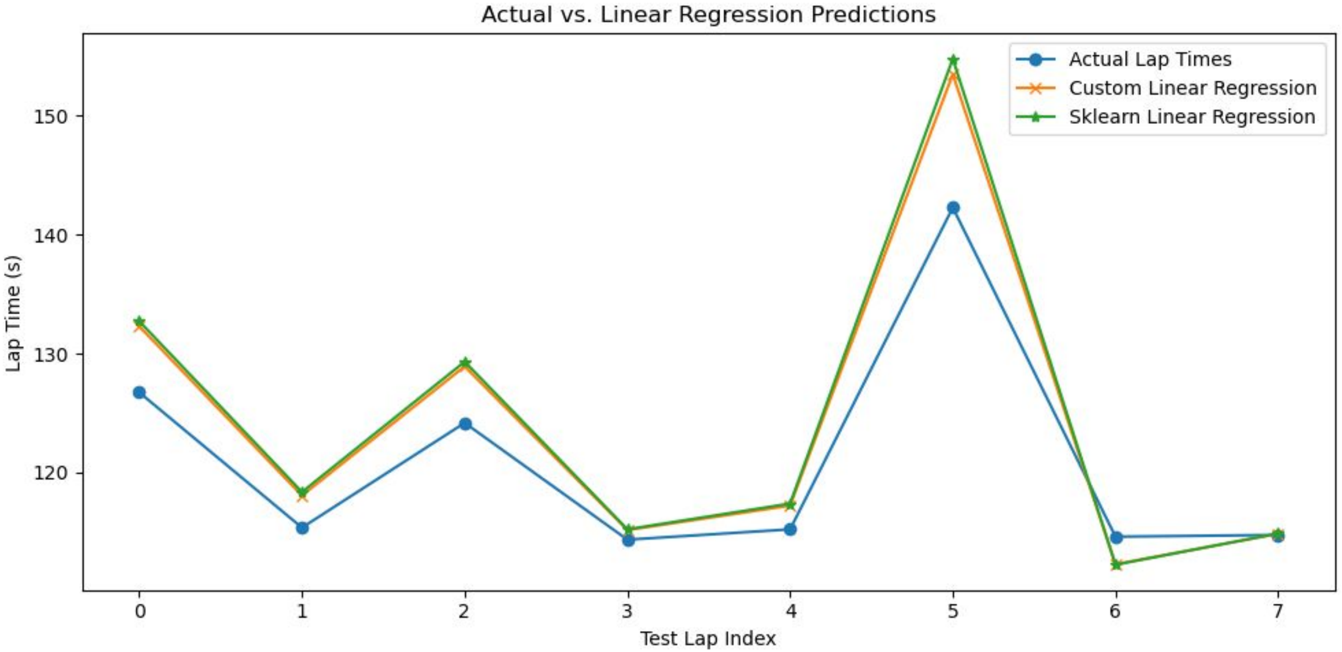
Linear Regression

Linear Regression Preparation

- ➢ Used Linear Regression to gather coefficients
- ➢ First try was using Normal Equation, but got thrown an error about Singular Matrix
  - ○ Fixed using Moore-Penrose Pseudoinverse
- ➢ Used scikit-learn Linear Regression model for comparison



Linear Regression Results



Sklearn Linear Regression MSE: 29.8143 | My Linear Regression MSE: 24.5447

Linear Regression Results

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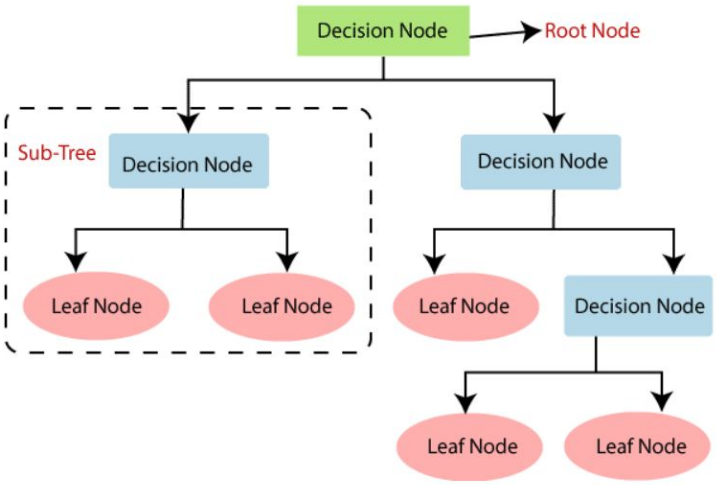
Coefficients:
Distance: -0.0001
LAP_BEACON: 0.0000
G_LAT: 0.0006
ROTY: 0.0060
STEERANGLE: -0.0655
SPEED: -0.2417
THROTTLE: -0.3248
BRAKE: 0.0488
GEAR: -0.0077
G_LON: 0.0025
CLUTCH: 0.1214
RPMS: -0.0194
TC: -0.0039
ABS: 0.0004
SUS_TRAVEL_LF: -0.0481
SUS_TRAVEL_RF: -0.0235
SUS_TRAVEL_LR: -0.0084
SUS_TRAVEL_RR: 0.0017
BRAKE_TEMP_LF: -0.2001
BRAKE_TEMP_RF: 0.0665
BRAKE_TEMP_LR: 0.0969
BRAKE_TEMP_RR: 0.0696
TYRE_PRESS_LF: 0.0233
TYRE_PRESS_RF: 0.0586
TYRE_PRESS_LR: 0.0116
TYRE_PRESS_RR: -0.0017
TYRE_TAIR_LF: 0.0556
TYRE_TAIR_RF: -0.0593
TYRE_TAIR_LR: -0.1349
TYRE_TAIR_RR: -0.1897
WHEEL_SPEED_LF: -0.0633
WHEEL_SPEED_RF: -0.0633
WHEEL_SPEED_LR: -0.0765
WHEEL_SPEED_RR: -0.0766
BUMPSTOPUP_RIDE_LF: -0.0047
BUMPSTOPUP_RIDE_RF: 0.0002
BUMPSTOPUP_RIDE_LR: 0.0000
BUMPSTOPUP_RIDE_RR: -0.0001
BUMPSTOPDN_RIDE_LF: -0.0002
BUMPSTOPDN_RIDE_RF: -0.0011
BUMPSTOPDN_RIDE_LR: -0.0004
BUMPSTOPDN_RIDE_RR: 0.0032
BUMPSTOP_FORCE_LF: 0.0960
BUMPSTOP_FORCE_RF: 0.0214
BUMPSTOP_FORCE_LR: -0.1368
BUMPSTOP_FORCE_RR: 0.1106
EN_ET: -0.0157
EN_TL: -0.0211
EN_TD: 0.0051
EN_TB: 0.0002
EN_TW: 0.0170
EN_W: 0.3516
EN_Dy: -0.0001
EN_AP: 0.0003
EN_GR: 0.0396
TIME: 0.2910
```

- ➢ Negative: SPEED, THROTTLE ⟶ faster lap times
- ➢ Positive surprises: BRAKE, ABS
- ➢ BRAKE_TEMP_LF ⟶ tighter left turns

Decision Trees

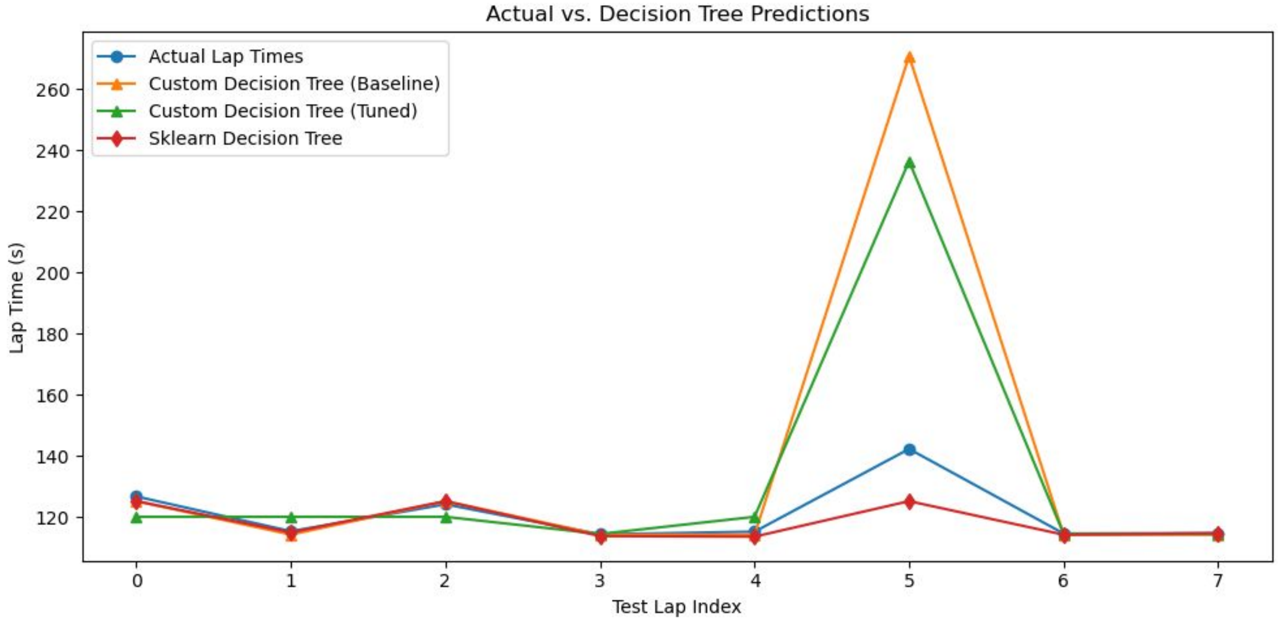
Decision Tree Preparation

- ➢ Built Decision Tree with the focus to minimize the MSE
- ➢ Initially overfit with huge errors (MSE ~2065)
- ➢ Hyperparameter tuning (Grid Search):
  - ○ max_depth: [None, 2, 3, 4, 5]
  - ○ min_samples_split: [2, 5, 10]
  - ○ min_samples_leaf: [1, 2, 5]
- ➢ Used scikit-learn Decision Tree model for comparison



Decision Tree Results

Best Parameters: max_depth=None, min_samples_split=2, min_samples_leaf=2



Sklearn Decision Tree MSE: 37.2577 | Base Decision Tree MSE: 2065.1080 | Tuned Decision Tree MSE: 1120.4481

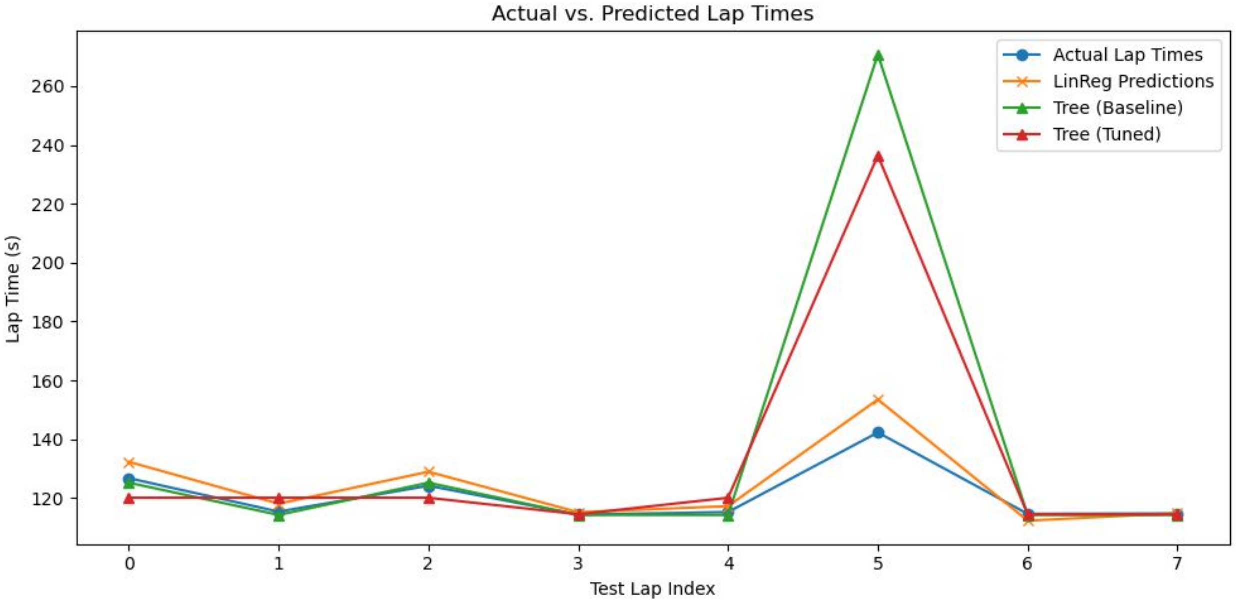
Decision Tree Results

Lap 0 | Actual: 126.74 | Tree(Base): 125.21 | Tree(Tuned): 120.09
Lap 1 | Actual: 115.37 | Tree(Base): 114.24 | Tree(Tuned): 120.09
Lap 2 | Actual: 124.18 | Tree(Base): 125.21 | Tree(Tuned): 120.09
Lap 3 | Actual: 114.36 | Tree(Base): 114.24 | Tree(Tuned): 114.47
Lap 4 | Actual: 115.23 | Tree(Base): 114.24 | Tree(Tuned): 120.09
Lap 5 | Actual: 142.28 | Tree(Base): 270.79 | Tree(Tuned): 236.39
Lap 6 | Actual: 114.61 | Tree(Base): 114.24 | Tree(Tuned): 114.47
Lap 7 | Actual: 114.75 | Tree(Base): 114.24 | Tree(Tuned): 114.47

- ➢ Tuned tree handled the lap with the spinout better
- ➢ All trees struggled on the outlier
- ➢ Tuned tree has smaller MSE, but is outperformed on every lap except the spin-out

Conclusion

Model Comparison



Model Comparison

Lap 0 | Actual: 126.74 | LinReg: 132.29 | Tree(Base): 125.21 | Tree(Tuned): 120.09
Lap 1 | Actual: 115.37 | LinReg: 118.07 | Tree(Base): 114.24 | Tree(Tuned): 120.09
Lap 2 | Actual: 124.18 | LinReg: 128.92 | Tree(Base): 125.21 | Tree(Tuned): 120.09
Lap 3 | Actual: 114.36 | LinReg: 115.17 | Tree(Base): 114.24 | Tree(Tuned): 114.47
Lap 4 | Actual: 115.23 | LinReg: 117.22 | Tree(Base): 114.24 | Tree(Tuned): 120.09
Lap 5 | Actual: 142.28 | LinReg: 153.49 | Tree(Base): 270.79 | Tree(Tuned): 236.39
Lap 6 | Actual: 114.61 | LinReg: 112.29 | Tree(Base): 114.24 | Tree(Tuned): 114.47
Lap 7 | Actual: 114.75 | LinReg: 114.86 | Tree(Base): 114.24 | Tree(Tuned): 114.47

Conclusions

- ➢ Linear Regression outperformed Decision Trees in terms of MSE
- ➢ Decision Trees outperformed Linear Regression in terms of accurately predicting lap times (6 out of 8)
- ➢ Neither of the models could accurately predict the spin-out lap, but Linear Regression did the best

Lessons Learned

- Use a larger dataset
- Use other methods to help the Decision Tree (Random Forests, etc)
- Choose the right metric
  - MSE does not necessarily mean best predicted lap time
  - Per-Lap Accuracy is a possible metric
- Maybe do not use any mistakes in future testing data

Questions?